- Change in breeding system.
- Increased yield.
- Increased resistance for disease and pest.
- Developing seedless parthenocarpic fruit.
- Enhancing colour, appearance, palatability and nutritional composition.

9.3 Origin of Agriculture

Archeological evidence for earliest record of agriculture is found in the fertile crescent region in and around Tigris and Euphrates river valleys, approximately about 12,000 years ago.

The earlier Greek and Roman naturalists like Theophrastus, Dioscorides, Pliny the elder and Galen laid down the scientific foundation in understanding origin and domestication of cultivated plants.

9.4 Organic Agriculture

Organic farming is an alternative agricultural system which originated early in the twentieth century in reaction to rapidly changing farming practices. It is a production system that sustains the health of the soils, ecosystems and people. It relies on ecological processes, biodiversity and cycles adapted to local conditions rather than the use of inputs with adverse effects.

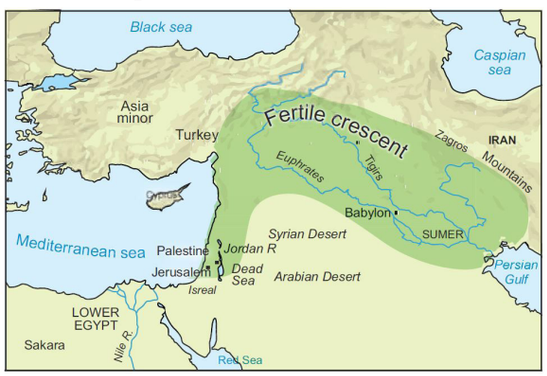


Figure 9.1 : Map shows Fertile crescent region

9.4.1. Biofertilizers

Biofertilizers are defined as preparations containing living cells or latent cells of efficient strains of microorganisms that help crop plants uptake of nutrients by their interactions in the rhizosphere when applied through seed or soil. Biofertilizers could be also called as microbial cultures, bioinoculants, bacterial inoculants or bacterial fertilizers.

They are efficient in fixing nitrogen, solubilising phosphate and decomposing cellulose. They are designed to improve the soil fertility, plant growth, and also the number and biological activity of beneficial microorganisms in the soil. They are eco-friendly organic agro inputs and are more efficient and cost effective than chemical fertilizers.



Indian Plant Breeders

a. **Dr. M. S. Swaminathan** – He is a pioneer mutation breeder.
b. **Sir. T.S. Venkataraman** – An eminent sugarcane breeder.
c. **Dr. B.P. Pal** – Famous wheat breeder, developed superior disease resistant varieties of wheat.
d. **Dr. K. Ramiah** – Eminent rice breeder, developed several high yielding varieties of rice.
e. **N.G.P. Rao** – An eminent sorghum breeder, developed world's first hybrid of Sorghum (CSH-1).
f. **C.T. Patel** – Who developed world's first cotton hybrid.
g. **Choudhary Ram Dhan** – Wheat breeder, who is famous for C-591 variety of wheat, which made Punjab the wheat granary of India.

S.N	Groups		Examples
A	**N_2 fixing Biofertilize**		
	1.	Free-living	*Azotobacter, Clostridium, Anabaena, Nostoc,*
	2.	Symbiotic	*Rhizobium, Anabaena azollae*
	3.	Associative Symbiotic	*Azospirillum*
B	**P Solubilizing Biofertilizer**		
	1.	Bacteria	*Bacillus subtilis, Pseudomonas striata*
	2.	Fungi	*Penicillium, Aspergillus.*
	3.	Arbuscular Mycorrhiza	*Glomus, Scutellospora.*
C	**P Mobilizing Biofertilizers**		
	1.	Ectomycorrhiza	*Amanita.*
D	**Biofertilizer for Micro nutrients**		
	1.	Silicate and Zinc solubilizers	*Bacillus.*
E	**Plant Growth Promoting Rhizobacteria**		
	1.	Pseudomonas	*Pseudomonas fluorescence*

Table 9.1: Classification of Biofertilizers

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